

2025 ISPE ANNUAL MEETING & EXPO

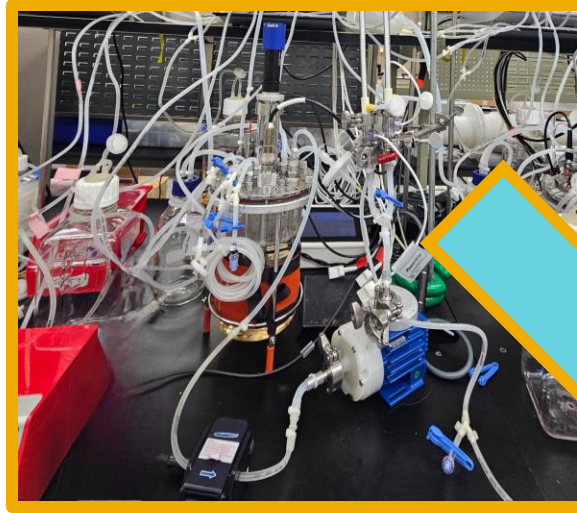
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Integrated Dynamic Bioprocessing Upstream Scale-up

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The information shown in this presentation is the outcome of an exhaustive effort over many years by countless scientists in AZ BPD. Key contributors to this specific effort of IDB are named and acknowledged. Those who are not: they know who they are, and know we could not have achieved upstream scale-up success without them 😊

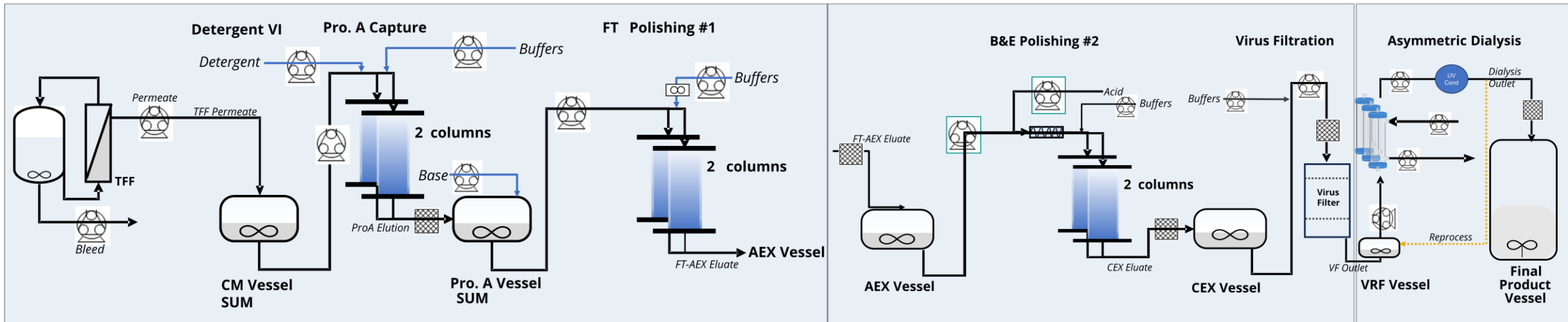
Integrated Dynamic Bioprocessing Upstream Scale-up

1. Defining Integrated Dynamic Bioprocessing (IDB)
2. Process and Equipment Design to Characterization, Commissioning, and Model building
3. GMP Implementation: People & Facility, Digitalization, and Quality
4. Process Monitoring and Troubleshooting

Defining Integrated Dynamic Bioprocessing (IDB)

What is Integrated Dynamic Bioprocessing (IDB)?

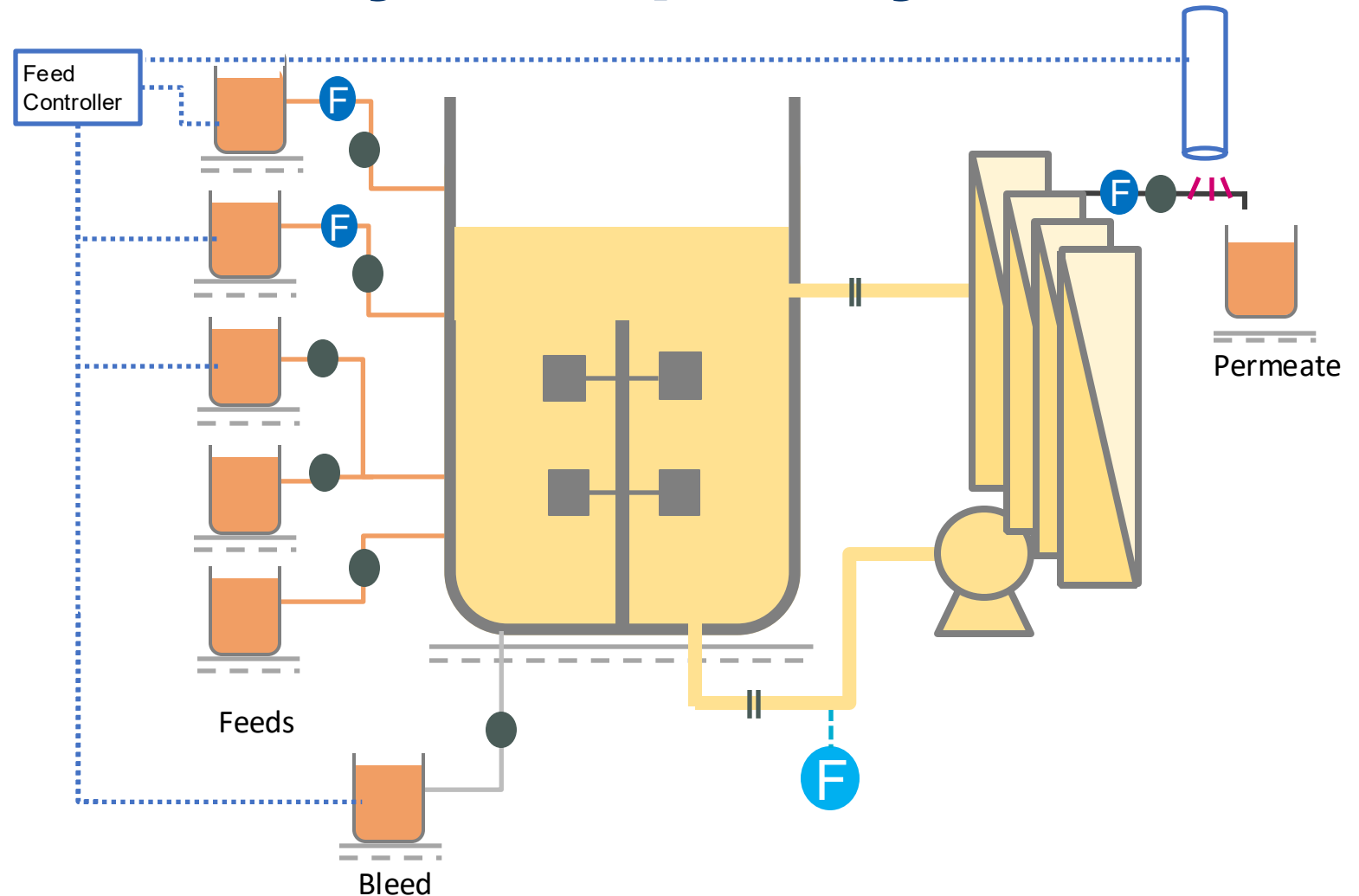
- A fully integrated, continuous, and adaptive process designed to increase upstream productivity and harvest of material five-fold as compared to fed-batch processing, using the same bioreactor footprint.
- **Bioprocessing** where up- and down- stream are seamlessly connected; upstream performance continuously informs downstream setpoints (i.e. **integrated**), run at high (> 100 M cells/mL) cell density, not at steady-state (i.e. **dynamic**)
- Goal: to implement in a **commercial** setting
- A process that's **intensified** beyond what off-the-shelf vendor solutions could support
- Utilizes **automation & PAT** feedback to support a dynamic process with minimal human interaction



Upstream IDB: Necessary Complexity

The novelty and uniqueness of upstream IDB is:

1. A **custom** bioreactor able to maintain high cell density mammalian cell culture
2. An automated feed controller with **PAT feedback for** continuous feeds and control of an optimal growth and production environment
3. **Tangential Flow Filtration (TFF)** and low-shear, magnetically levitated pump, where the circulation of cells is optimized to minimize shear and residence time to minimize cell stress.
4. Unique **TFF skid** design and control strategies to minimize filter area requirements



Process and Equipment Design to Characterization, Commissioning & Model Building

Out of the frying pan (covid-19) and into the fire

- When IDB scale-up began in earnest in 2021, there were **no robust**, off-the-shelf single-use (SU) bioreactor solutions for the IDB requirements (e.g. high cell density + TFF integration)
- Legacy stainless and SU bioreactors were tested, and pushed to their **limits**, resulting in ruptured sparge tubing, TFF pump issues, and fouled exhaust filters
- Residual **supply chain delays** from the covid-19 pandemic significantly impacted 50 L trials
- A bioreactor vendor permitting **customizability** was chosen, and a custom bioreactor was designed to meet the requirements of IDB upstream scale-up



Designing a Scale-up IDB Bioreactor

To support the higher cell densities (>100 M cells/mL), bioreactor design elements including:

- Higher oxygen transfer rate ($k_L a$ up to 30 1/hr)
- Automated controls for stripping of accumulated CO_2 and antifoam addition
- Exhaust management to accommodate the high gas flow rates
- Ability to integrate multiple TFFs with desirable ports locations to minimize gas entrainment into the TFF and to minimize pump speed (i.e. cell shearing)



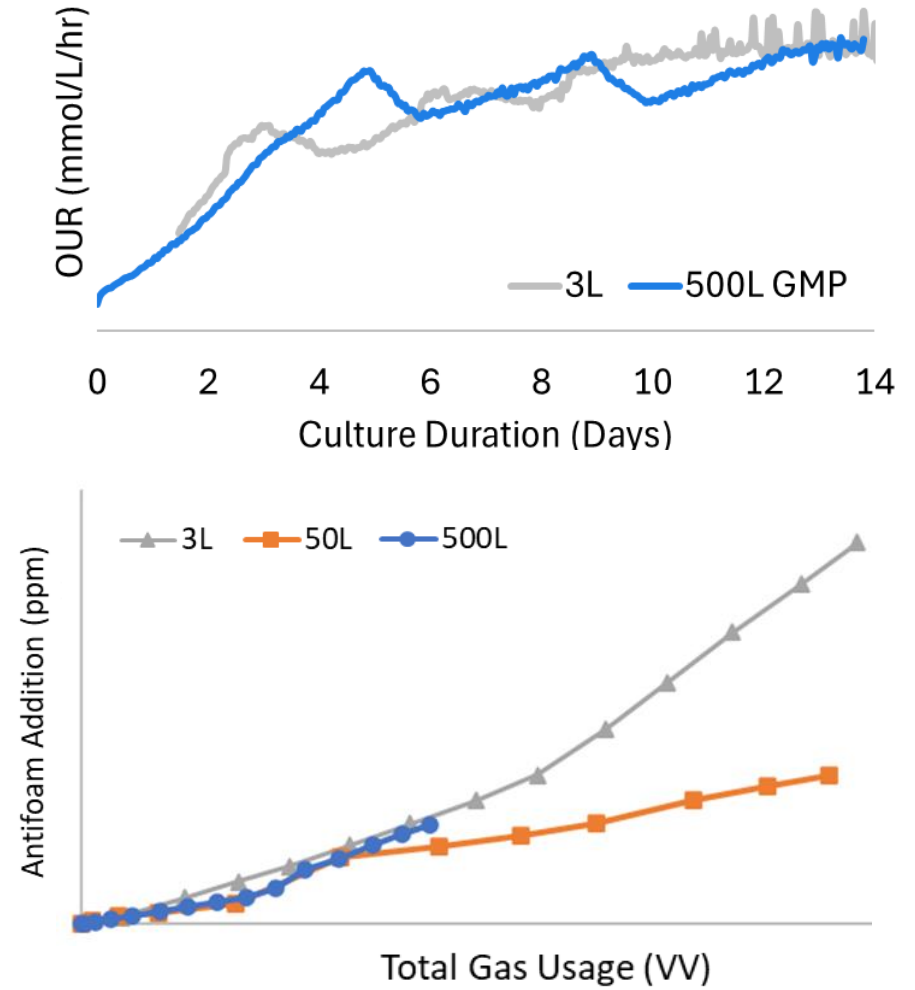
Scaling up from the Bench

Small/mid-scale Oxygen uptake Rate (**OUR**) calculations are used to:

- Assess process **scalability** and determine operating region for large scale runs (including predicting O_2 flow rate and agitation profiles)

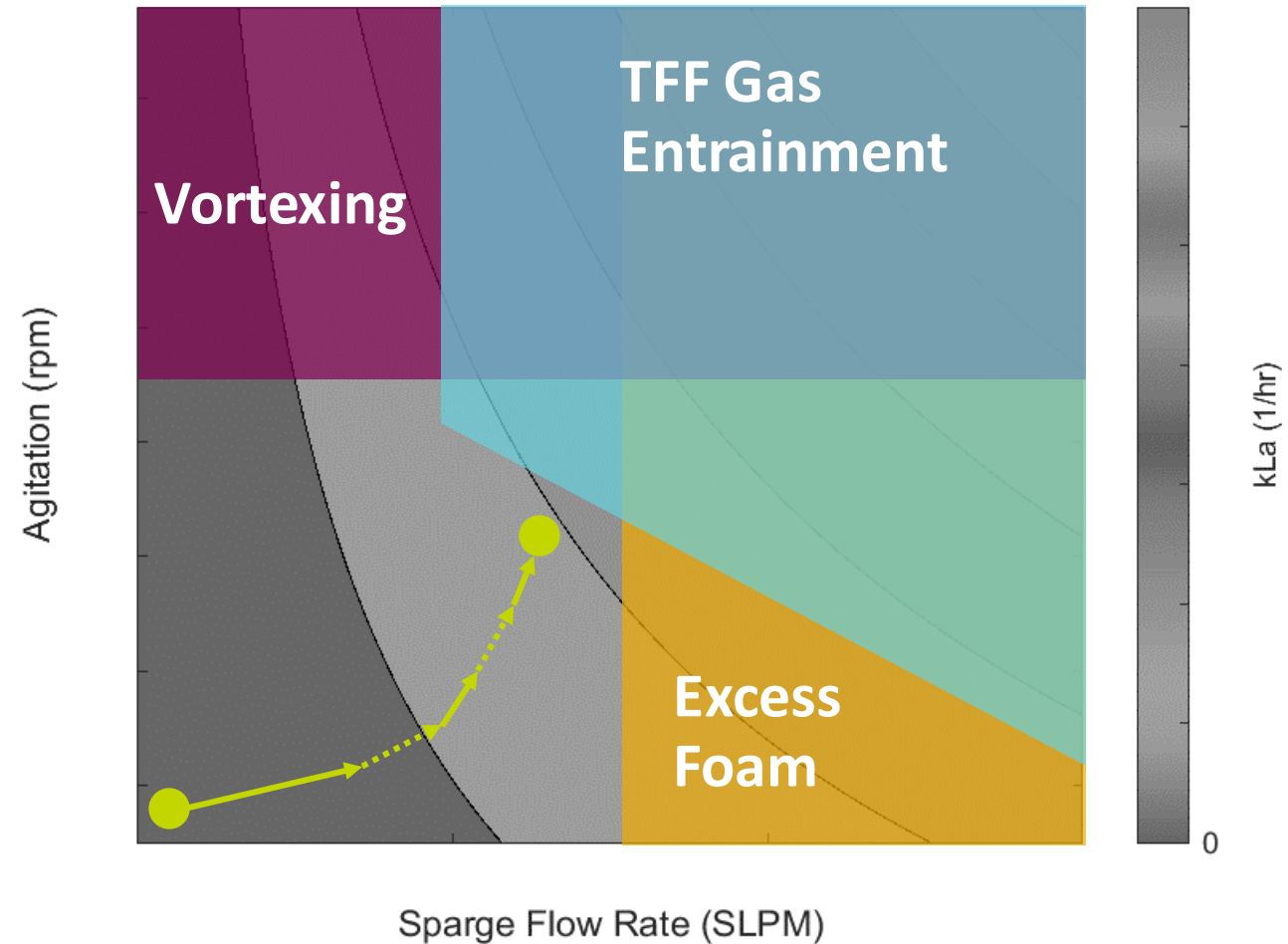
For IDB scale-up we:

- Design a DO control strategy: **DO profile ramping** – removes operator interference to manually step up over the wide range of required O_2 mass transfer
- Design a **pCO₂ control air cascade** strategy where air sparge is proportional to pH (when pH drops, CO₂ stripping increases)
- Design an **automated antifoam** strategy where antifoam addition is proportional to gas flow rate and agitation

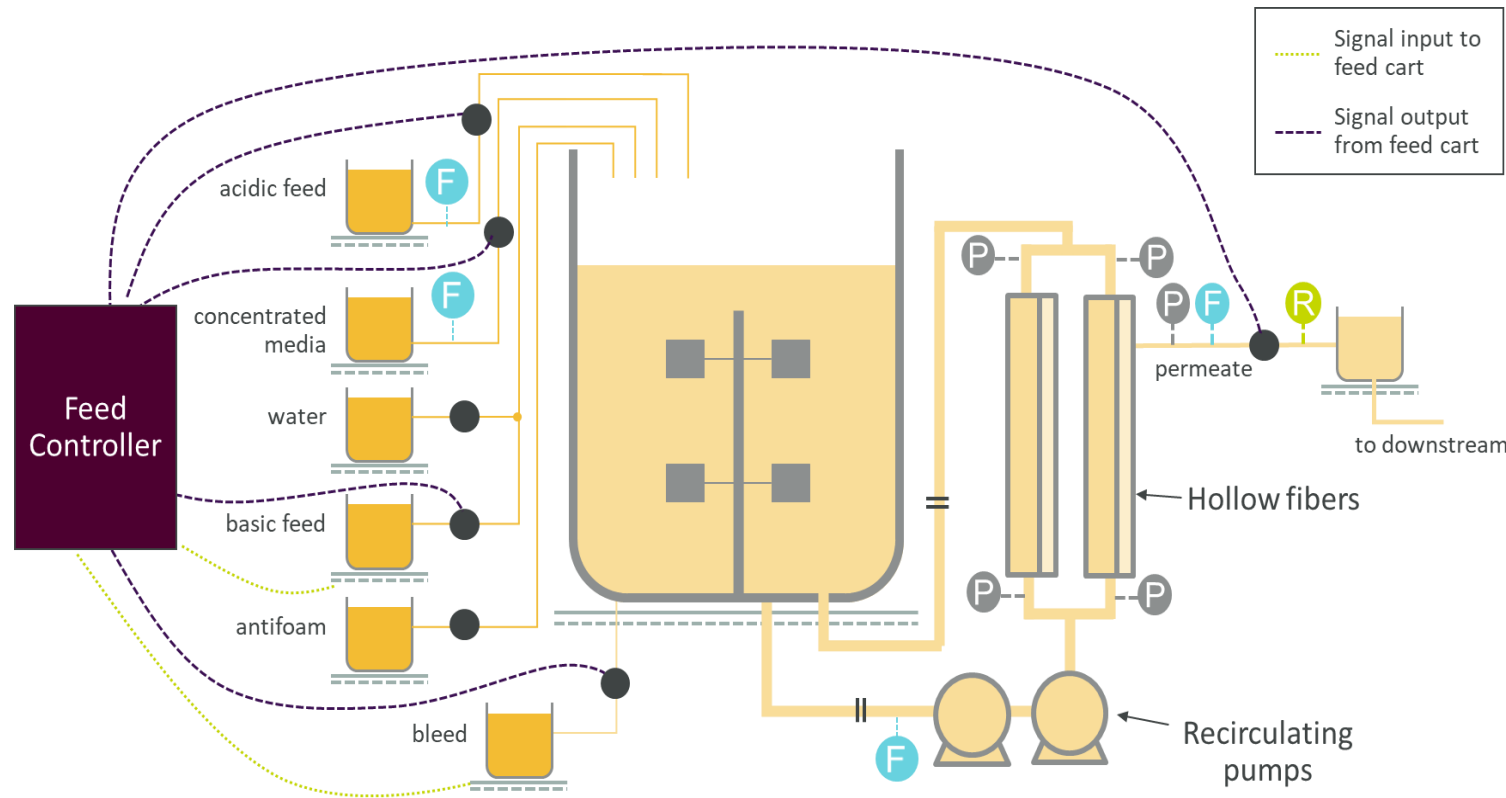


Achieving Stability, Creating Confidence

- Scale-up IDB bioreactor must meet the classic bioreactor **requirements** of sufficient mixing and mass transfer while avoiding excess foaming and acidic CO₂ accumulation
- An additional degree of restraint is added, where gas bubble **entrainment** into the TFF recirculation pump must be reduced
- The yellow path represents an ideal **operating strategy**, where sparging and agitation are increased in turn to move along the mass transfer (kLa) contours, while avoiding the **deleterious** phenomena
- The final key is to **automate** this strategy to avoid the need for manual operator decision making and intervention



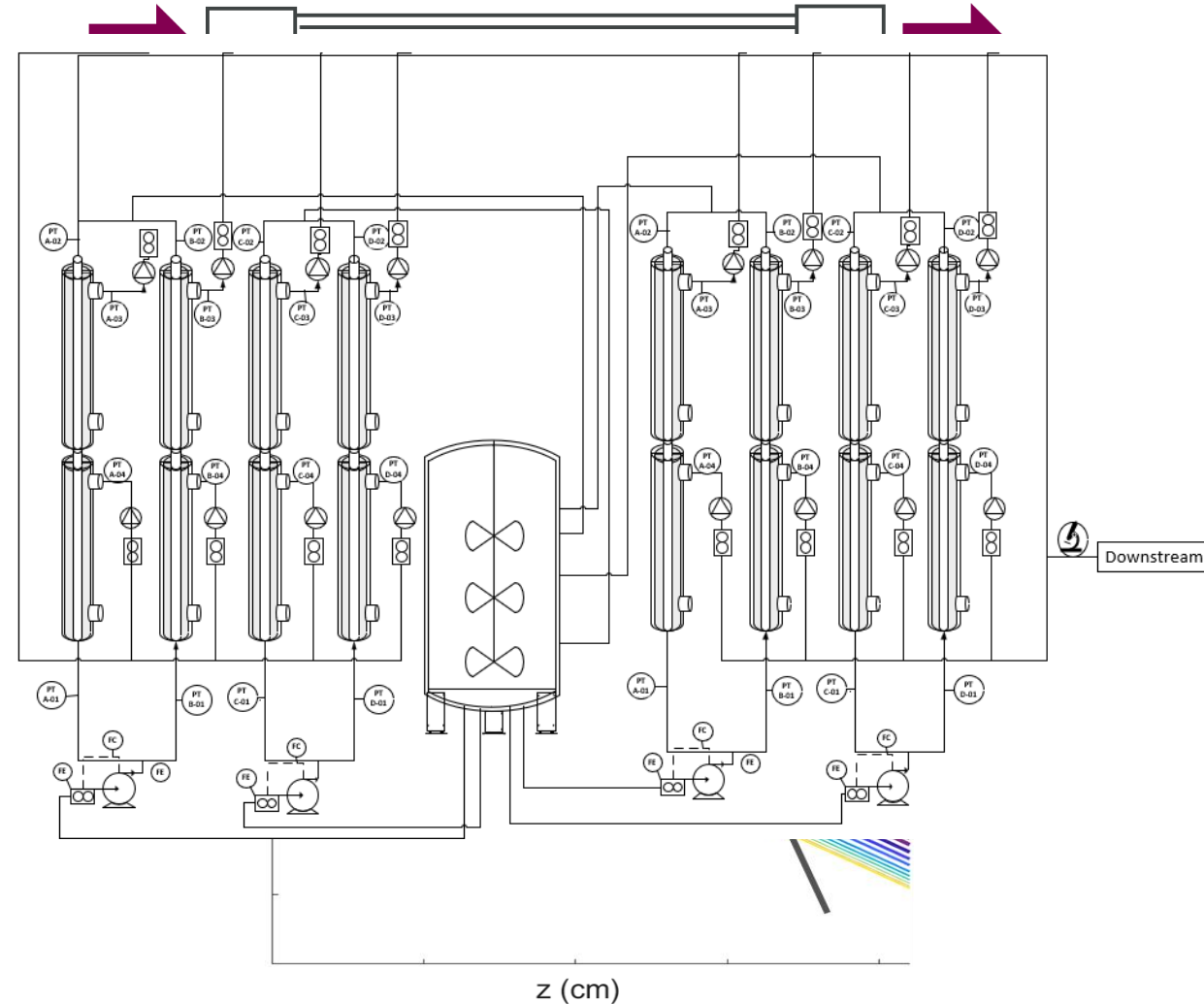
Automated Feed & Bioreactor Bleed Controller



- A recipe-based automated feed controller with PAT feedback for continuous feeding to maintain a glucose concentration
- Raman in the permeate stream for ease of measurement and adaptive feedback control for variable glucose consumption rates
- Gravimetric and volumetric control of inlet and outlet streams using scales and flowmeters
- Concentrated media and feeds to reduce volume requirements

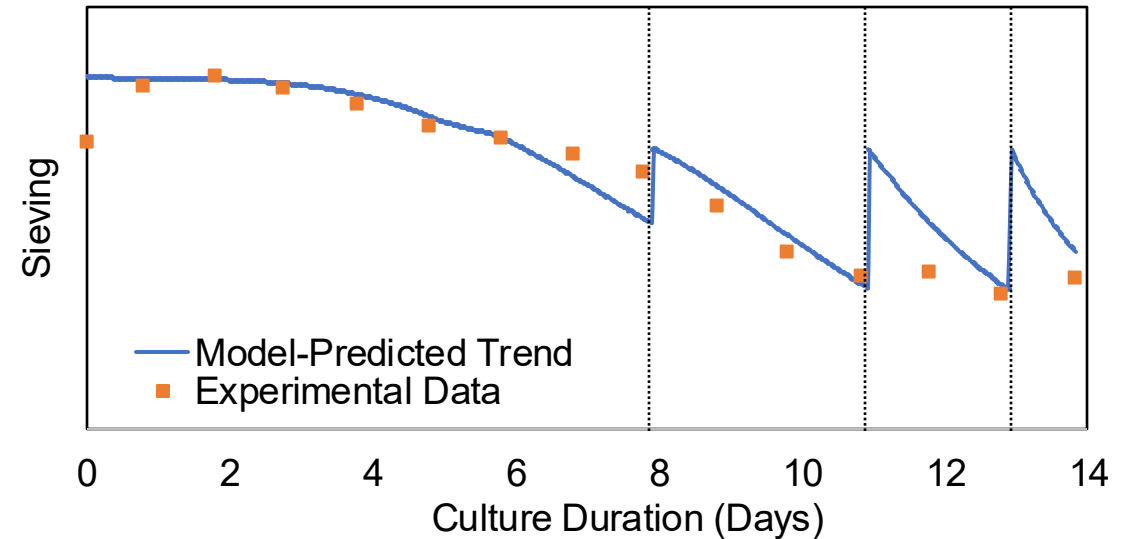
Scaling Tangential Flow Filtration

- TFF was chosen over ATF for **scalability** to 2 kL+ and better control at high culture **viscosity**
- Scaling the bench platform would require **16+ TFFs** operating simultaneously for a single 2 kL bioreactor!
 - Too many **ports** on bioreactor bag and pumps, enormous facility **footprint**, and significant demand on **operators**
 - Not practical for facility fit!
- Engineering solutions to reduce filter requirements:
 - Advantageous filter **geometries** – reduces the ΔP across the fiber length – limited vendors...
 - Increase flux for higher throughput → operate filters **sequentially**, swap filters when fouled



TFF Scale-up: More than 1 Filter is Needed!!!

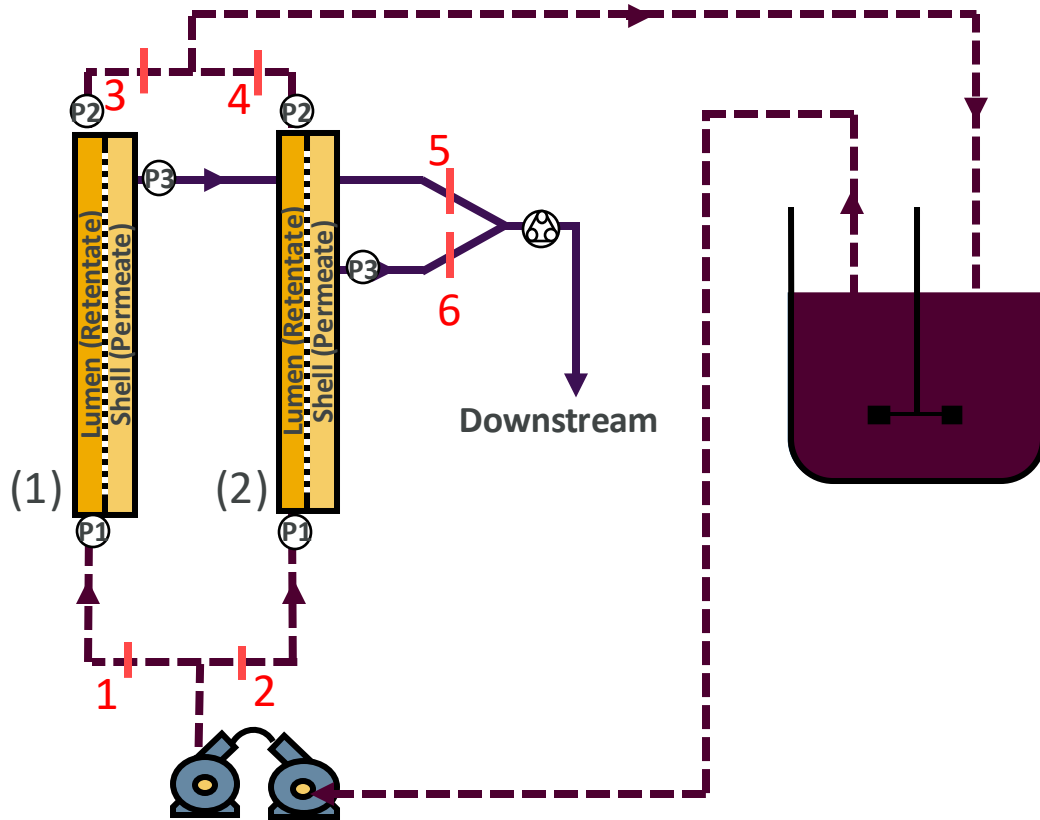
- Filters **foul** due to cell debris buildup; multiple filters are needed to complete runs
- Clinical and commercial IDB requires TFF filter **swaps** (dashed lines) when sieving diminishes
- Sieving is a function of the cell culture content – dirtier, more viscous culture → lower sieving → more frequent filter swaps
- Predicting **optimal** swapping time is paramount to avoid critical filter fouling
- TFF swapping is **not a trivial** procedure with viscous high cell density culture; it requires careful setup, coordination, equipment usage, and understanding of fluid dynamics
- The TFF swapping procedure was the **most adapted** and modified part of IDB upstream scale-up as the challenges were scale-specific



Filter swaps pre-planned for days 8, 11, and 13 based on **model predictions** from historical, small scale data (goal: **maximize** filter use and avoid low sieving to provide downstream with consistently high concentrations of product)

Since this initial 500L run, the swap procedure has been further optimized to improve post-swap sieving and the model have been refined to better represent the expected sieving post-swap

TFF Reverse Flow, Regen, and Automation

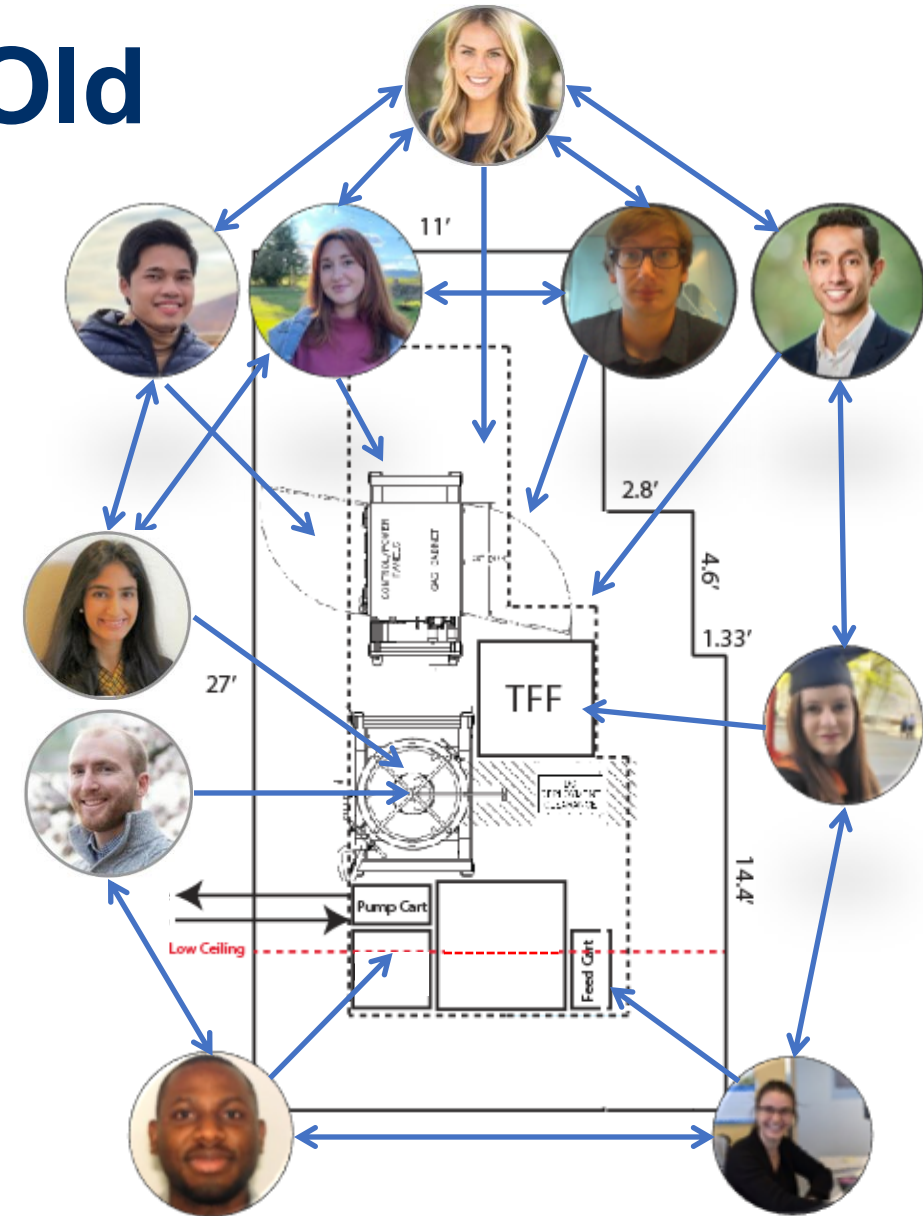


- TFF control and optimization is at the very heart of IDB scale-up: reduce **fouling**, maximize **sieving**
- To leverage the entire usable filter area, cell culture flow direction **switches**, mitigating Starling Flow (i.e. 'reverse flow' operation)
- Additionally, TFF filters are **regenerated** using NaOH to remove foulants and extend filter lifetime (Ana Dilillo, et al., BPI East, 2025).
- Key to reversible flow, swapping and regeneration is **valve** and **pump** orchestration to sufficiently clean the filter, avoid NaOH contact with cell culture, and to maintain recirculation.
- To avoid operator decision making and intervention, these processes are now **automated** 😊

GMP Implementation: People & Facility, Digitalization, and Quality

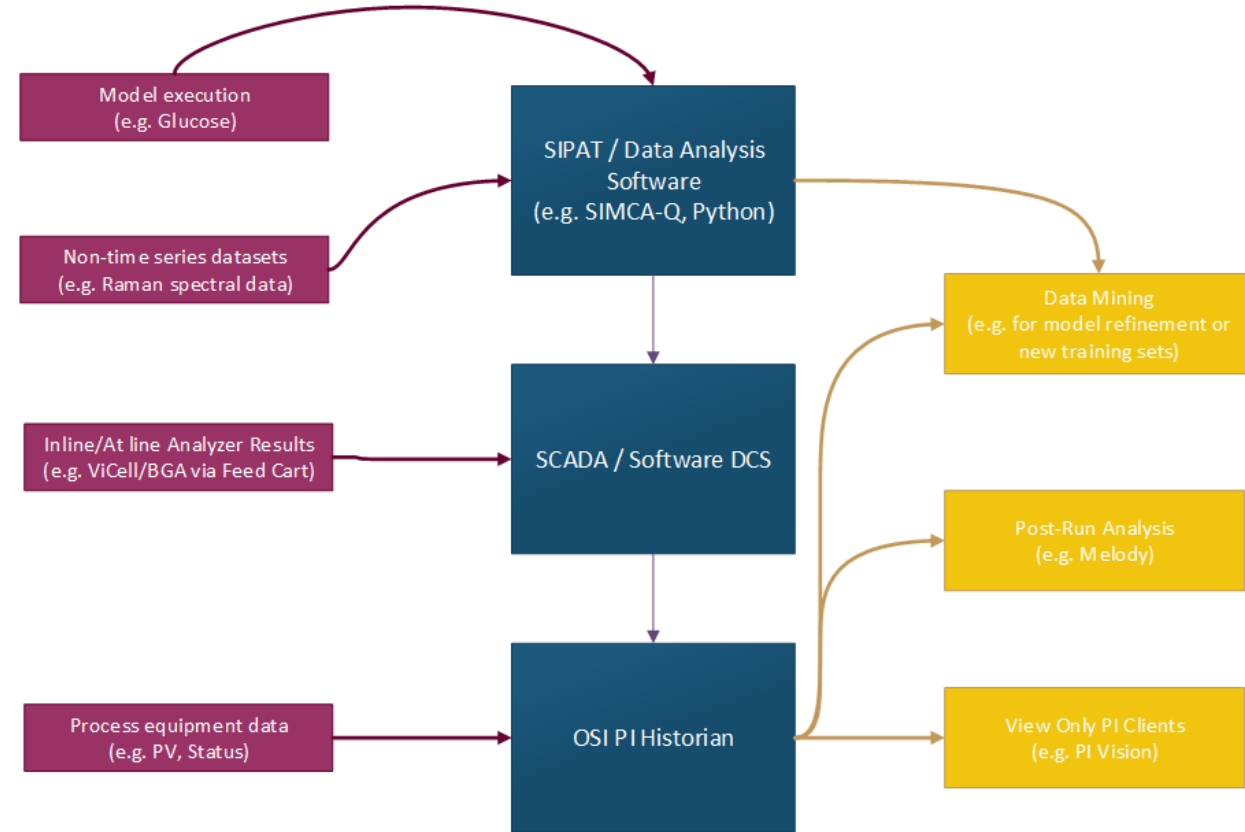
People and Facility: New and Old

- IDB was **novel** to the *existing* GMP manufacturing in numerous ways: a complex process, new unit operations (TFF for upstream, automated feed controller, Raman spectroscopy), and single-use technology
- Both **physical** and **human** factors were key: adapting technology to the facility, aligning utilities, training personal
- Issues included significant space limitations, utilities limitations, and a host of new vendors and part numbers – **procurement**
- A **cross-functional** team was required from development, through manufacturing, procurement, and quality: the complexity and novelty was adapted for many different **ways of thinking**
- **Communication** between all areas is key (while not having infinite meetings): upstream to downstream, timelines to procurement, safety to quality; procedure and data must be properly managed while working on aggressive timelines



Digital Implementation: Data Management

- Upstream is historically the less **complex** side of drug substance: fewer unit ops than downstream
- IDB requires more unit ops (BRX, feed tanks, TFF); the feed cart and algorithm requires more **data types** (offline data, online data, Raman data)
- Unit operations for IDB are **interconnected**: equipment status and data are paramount
- Unit operations have their own data formats: **standardize** for storage and retrieval
- IDB also requires standardization of **access** to data, model **version** and control management, and **contextualization** of raw data AND analyses
- Want to create an infrastructure where additional models and automation can be incorporated in the future



Training and Technology: Ensuring Quality

Training

- IDB was novel to the GMP facility
- Single-use technology was novel too
- Hands-on training by SMEs and vendors
- SOP content generated from training
- Pictures and videos!!!

Global Change Control

- Change management: GMP facility
- SOPs and MPRs: drafting and approval routing while the process was being developed
- Physical systems, automation, and data management systems
- Significantly more documentation than fed-batch!!!

Risk-based Validation

- Quality by design: follow the URS (all of them)
- IQ/OQ/Validation
- Compliant with standard processes
- Novelty of IDB technology!!!

Communication

- Key representatives reduce meeting headcount, but led to misunderstandings
- Key concerns were missed without all SMEs present and contributing
- Duplicated work became an issue without oversight and alignment

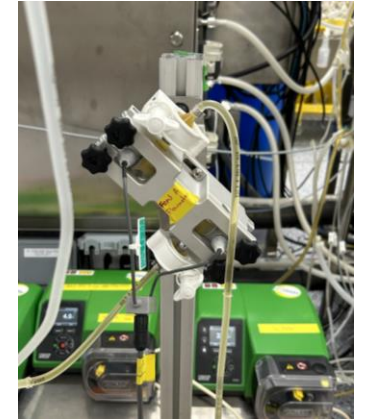
Process Monitoring & Troubleshooting

A Plethora of Failure Modes

- Single-use technology new to the GMP facility!!!
- Issues and challenges in development
 - Numerous **sterile connections** (>> 20 !!!)
 - **TFF pump** stalls and failures during TFF swapping
 - Strategy to monitor **TFF fouling** over 14 days
 - Complex **feeding algorithms**, user input sensitivity
 - **Accuracy** of sensors (flow meter + scales)
 - **Leaks** on gamma assemblies during pressure tests
 - Minimal **model** parameterization data
 - Lack of physical **flexibility** (port options)

Outcome: Failure Mode and Effects analysis (FMEA)

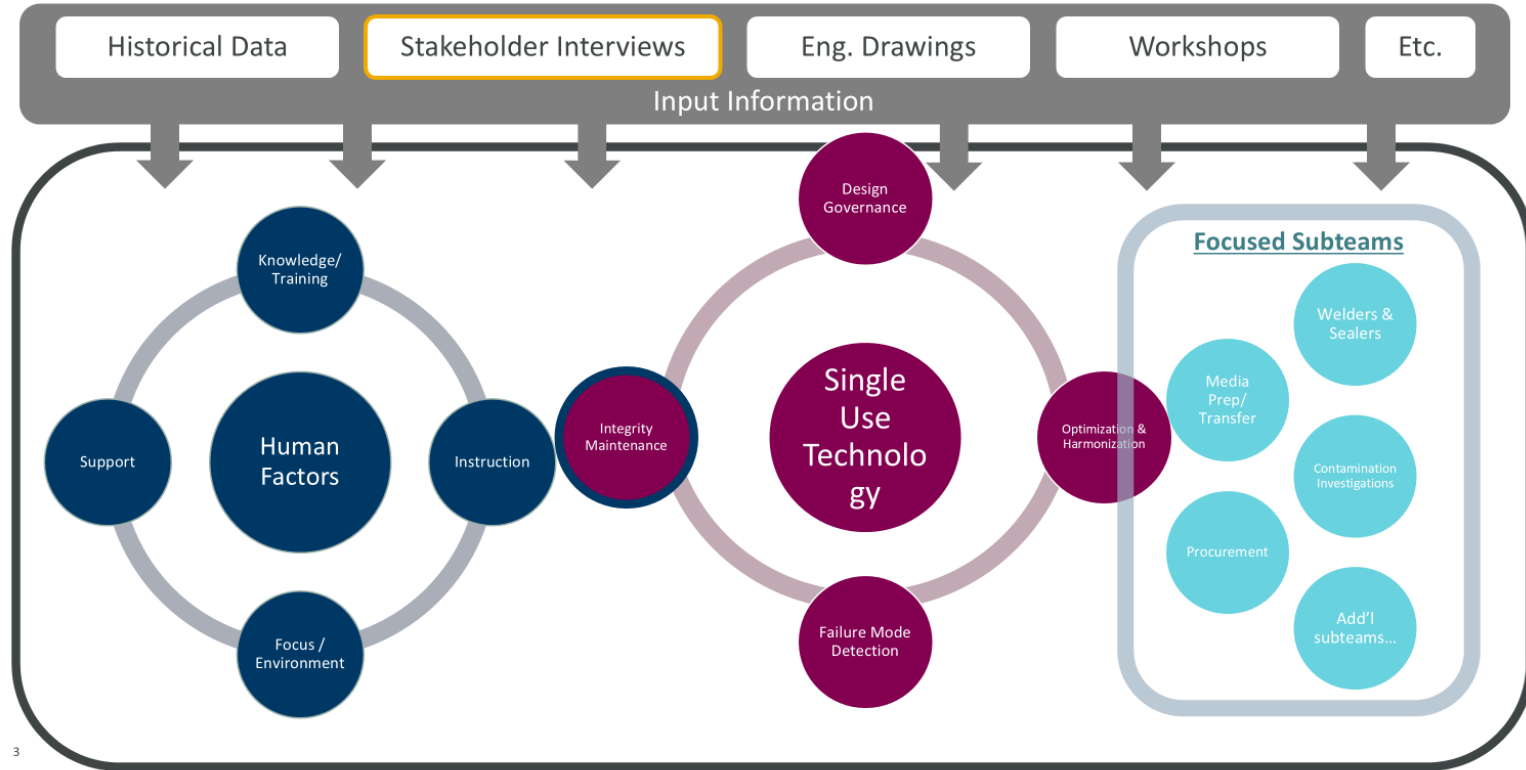
- Prior to **GMP** implementation
- Address **quality** concerns *proactively*
- Criticality of **alarms**



Proper Supervision, Constant Vigilance

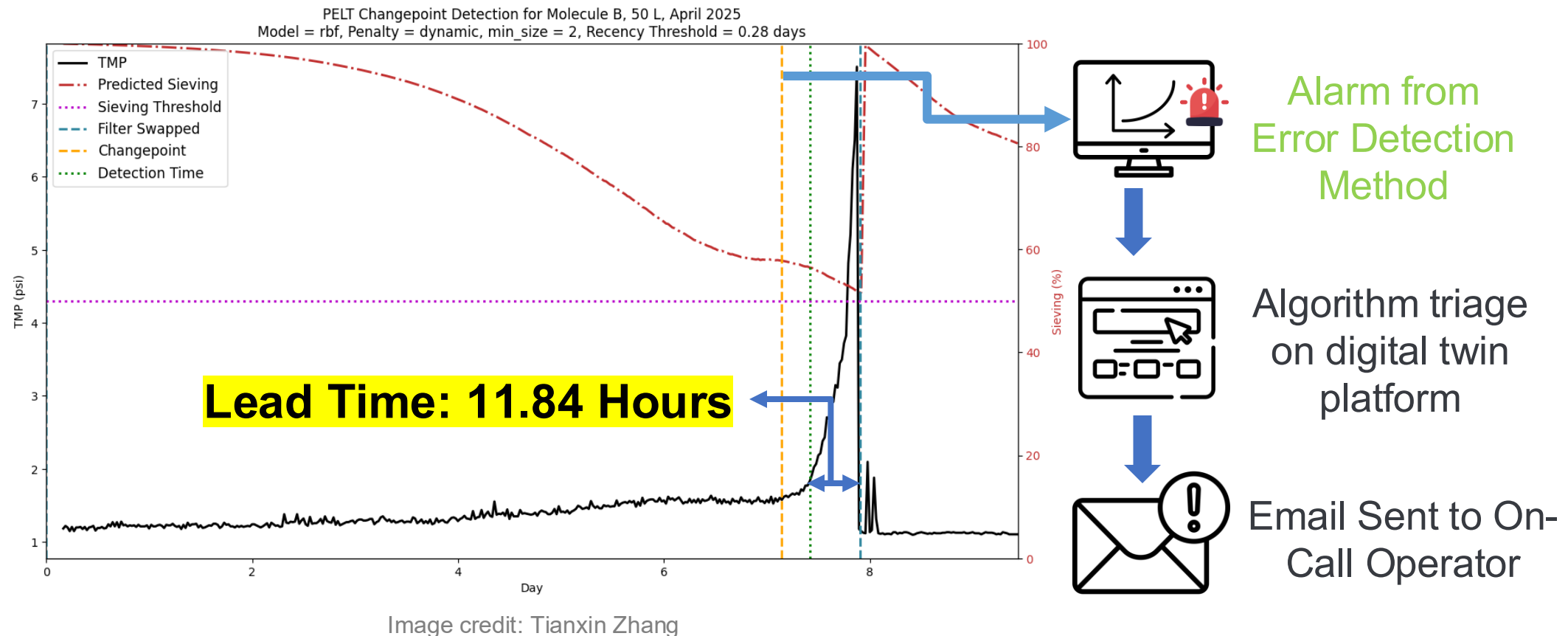
Contamination issues led to the formation of a 'sterile envelop team'

- Split into human factors and single-use technology
- Focused sub-teams generated detailed recommendations:
- Importance of adequate time for run preparation
- **Inspect, test, and compare** all equipment to pre-defined specifications
- Leverage all **training** materials
- Use **uniform** OSI-PI dashboards for easy, remote data monitoring
- Configure **alarms** appropriately to provide warnings for adequate time to intervene
- Around the clock supervision: what if a **digital twin** could be leveraged?



Next Generation Monitoring: Digital Twins

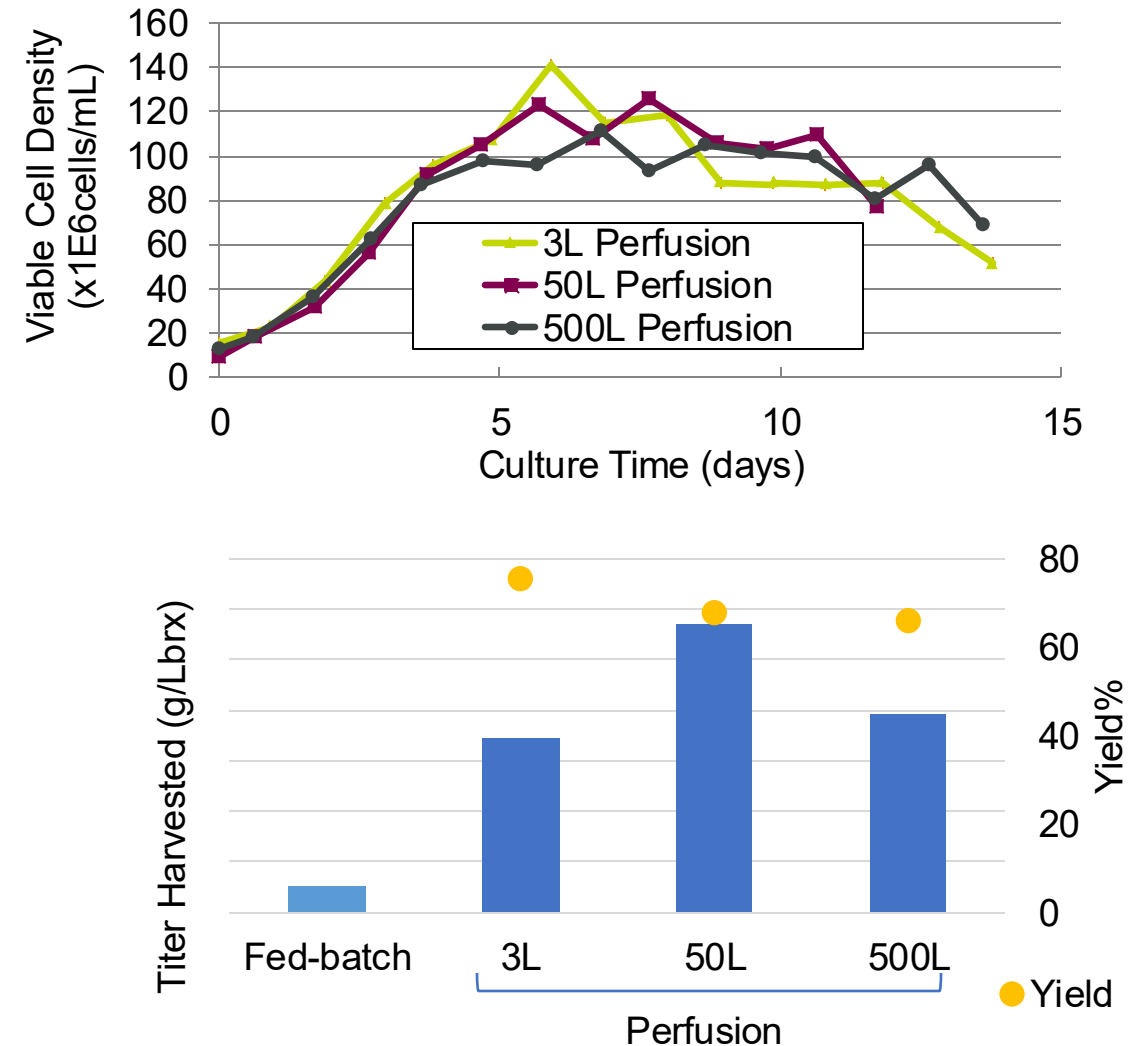
- **Digital Twins** can provide early error detections for operators on decision making
- Change point detection (PELT) for filter swap **early detections**



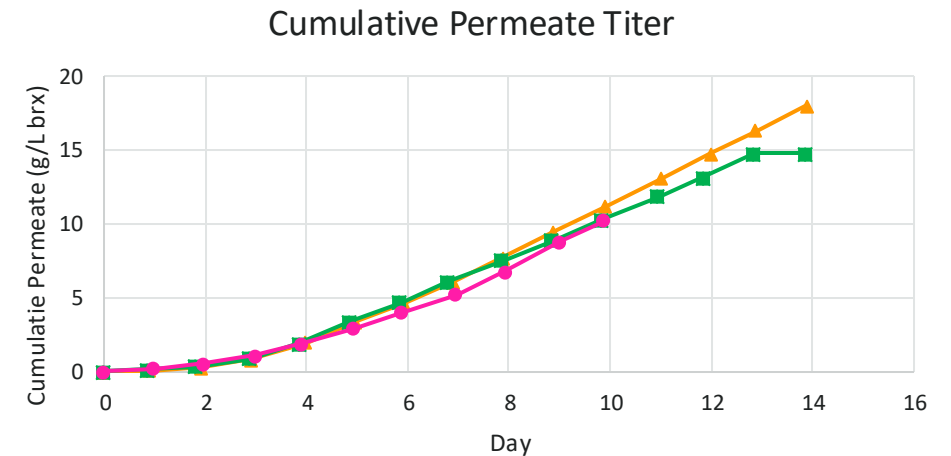
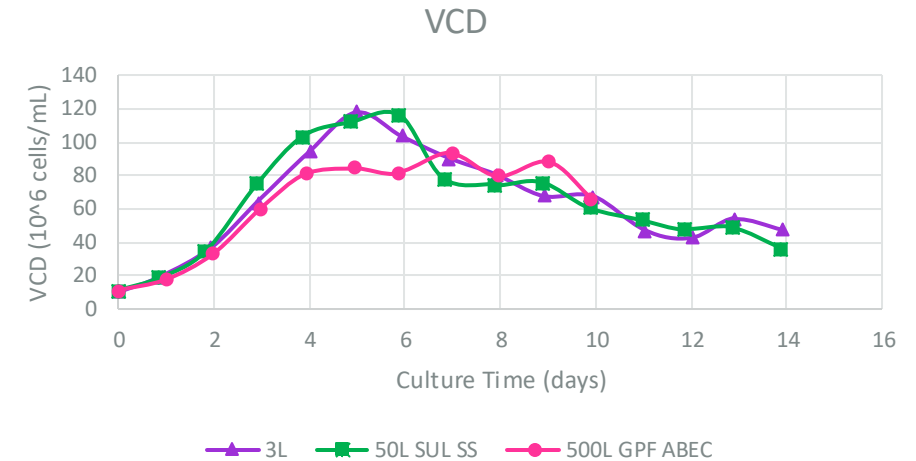
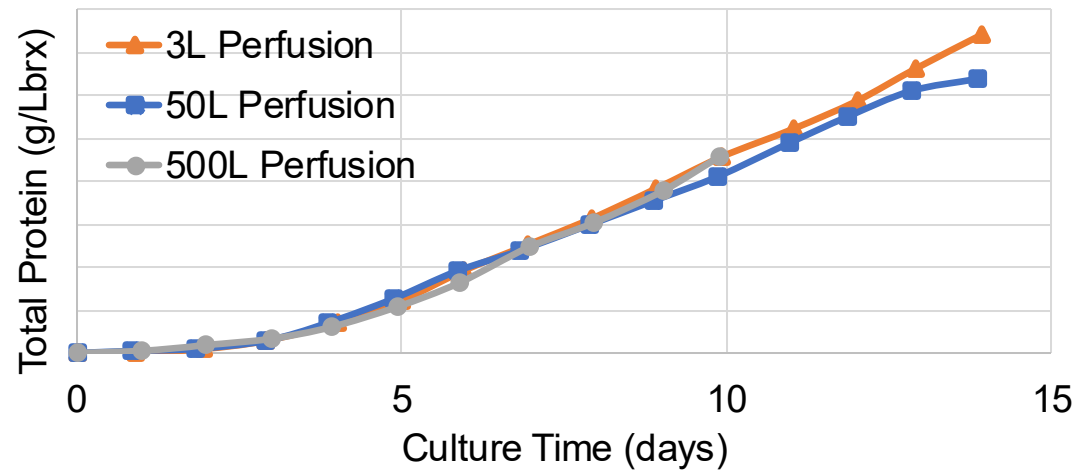
Conclusion and Future Steps

Sweet, Sweet Victory

- 500L GLP Tox run in the development space with Molecule #1
 - On average across scales, **~9.3x more product was harvested as compared to the original fed-batch process**
- Productivity improvements achieved at the 500 L scale with full GMP compliance
 - x3 successful lots, x2 products (Molecule #2 data on next slide)
- Minor deviations occurred and were triaged with quality to permit the release of **clinical material**
- The **legacy** Gaithersburg Pilot Facility (GPF) was successfully adapted and equipment allocated and integrated to permit novel IDB
- **Product quality** was also maintained across scale-up, and from fed-batch



GMP Engineering Run with Molecule #2



Conclusion and Next Steps

- It's not just the process, it's the **people**
- **Communication** is key
- **Resource planning** and sharing is essential
- The collection, processing, sharing, and storing of **data** is paramount
- Flying is plane while building the wings isn't trivial, but it is possible (please do not actually try this)

For IDB to be economically viable, the goal is to **commercialize** the process up to 10 kL or even 20 kL total bioreactor volume of IDB (12 or 24 kL of permeated per day!!!)



Image credit: T.M. Duncan

Acknowledgments

- Bioprocess technologies and Engineering (BTE)
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- Manufacturing Technology and Development (MTD)
- GMP Manufacturing Sciences (MS)
- Engineering Technical Services (ETS)
- Automation Engineering (AE)
- Procurement
- Quality Assurance (QA)
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